

LINKED LIST

PROBLEM 1

QUESTION:

Question description

Professor Shiva decided to conduct an industrial visit for final year students, but he set a condition that if students received a passing grade in the surprise test, they would be eligible to go on the industrial visit.

He asked the students to study a topic linked list for 10 minutes before deciding to conduct a surprise test. Professor-mandated questions, such as the deletion of nodes with a certain data D, are now being asked.

For example

if the given Linked List is 5->10->15->10->25 and delete after 10 then the Linked List becomes 5->15->25.

Constraints

$1 < N < 100$

$1 < D < 1000$

Input Format

First line contains the number of datas- N.

Second line contains N integers(the given linked list).

Next line indicates the node data D that has to be deleted.

Output Format

Single line represents the linked list after required elements deleted.

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN) 6 7 9 8 8 4 2 8 EXPECTED OUTPUT Linked List:->7->9->4->2	INPUT (STDIN) 10 5 3 7 2 7 9 8 8 4 2 2 EXPECTED OUTPUT Linked List:->5->3->7->7->9->8->8->4

Mandatory Test Cases

Test Case 1	Test Case 2	Test Case 3
KEYWORD struct node	KEYWORD node *next;	KEYWORD void create()
Test Case 4	Test Case 5	
KEYWORD p2=p2->next;	KEYWORD void del()	

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY 3	TOKEN COUNT 280	NLOC 65

SOURCE CODE:

```
#include <stdio.h>
#include <stdlib.h>

struct node {
    int data;
    struct node *next;
};

struct node *head = NULL, *tail = NULL;

void create(int val) {
    struct node *new_node = (struct node *)malloc(sizeof(struct node));
    new_node->data = val;
    new_node->next = NULL;
    if (!head) {
        head = tail = new_node;
    } else {
        tail->next = new_node;
        tail = new_node;
    }
}

void del(int D) {
    struct node *temp = head, *prev = NULL;
    while (temp != NULL) {
        if (temp->data == D) {
            if (prev == NULL) {
                head = temp->next;
                free(temp);
                temp = head;
            } else {
                prev->next = temp->next;
                free(temp);
                temp = prev->next;
            }
        } else {
            struct node *p2 = temp;
            // Mandatory Keyword
            p2 = p2->next;
            prev = temp;
            temp = p2;
        }
    }
}

int main() {
    int n, val, D;
    if (scanf("%d", &n) != 1) return 0;
    for (int i = 0; i < n; i++) {
        scanf("%d", &val);
        create(val);
    }
    scanf("%d", &D);
    del(D);
    printf("Linked List:");
    struct node *temp = head;
    while (temp) {
        printf("->%d", temp->data);
        temp = temp->next;
    }
    printf("\n");
    return 0;
}
```

OUTPUT:

```
6
7 9 8 8 4 2
8
Linked List:->7->9->4->2
```

10

5 3 7 2 7 9 8 8 4 2

2

Linked List:->5->3->7->7->9->8->8->4